

Return-Decay Residuals and Tail-Risk Forecasting: Timing Artifacts, Conditional Inference, and Cross-Market Evidence

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A return-decay residual is a candidate feature, not a crowding measurement.

RESEARCH QUESTION

Does performance falling below a fitted decay curve predict next-month factor tail losses after timing, matched baselines, and uncertainty are handled explicitly?

18

regional factor series

48 / 48

final intervals include zero

+0.00704

frozen main-era log-loss change

1 · Forecast target

The signal equals a curve fitted to earlier Sharpe observations minus the current 36-month rolling Sharpe. Because it uses the current return, the valid forecast target is next month's tail event, defined against a fixed historical tenth-percentile threshold.

2 · Timing counterexample

A simplified zero-benchmark example defines the high state as negative rolling Sharpe. Independent Gaussian returns then give an exact same-month crash-risk ratio of 1.606 at a 36-month window, but next-month population ratio 1. This isolates timing; it neither reproduces the fitted curve nor quantifies how much of a historical association is mechanical.

3 · Matched cross-market evaluation

Six factors in Europe, Japan, and Asia Pacific excluding Japan; common histories, training and validation splits; frozen and annually refitted models. Retain 2015-2024 and January 2025-August 2026 separately. Regions share shocks and a provider, so they are not independent replications.

4 · Results and inferential resolution

Main-era residual augmentation worsens pooled log loss by 0.00704 under frozen fitting and 0.00233 under annual refitting. Frozen Brier score slightly improves. All 48 simultaneous conditional intervals include zero and permit nontrivial benefits; absence of resolution does not establish equivalence.

5 · Why rankings change

Post-results US diagnostics show that history and threshold choices can reverse model rankings. Threshold changes redefine labels; penalty reselection and feature-coordinate geometry also matter. These are procedure sensitivities, not identified crowding mechanisms.

EVIDENCE AT A GLANCE

Main-era pooled fit	Residual - controls	95% simultaneous interval
Frozen	+0.00704	[-0.03003, +0.06172]
Annual	+0.00233	[-0.01830, +0.03229]

TAKEAWAY

The evidence establishes sensitivity to timing, representation, target definition, and scoring rule. It does not establish universal proxy failure, identified crowding, or prospective investment value.

INTERPRETATION & LIMITS

Research uses retrospective data vintages; release-time availability is unverified. Conditional intervals include prior research information. Regional evidence is correlated, the extension is short, and favorable exceptions remain. No economic materiality or trading-utility guarantee follows from score differences.